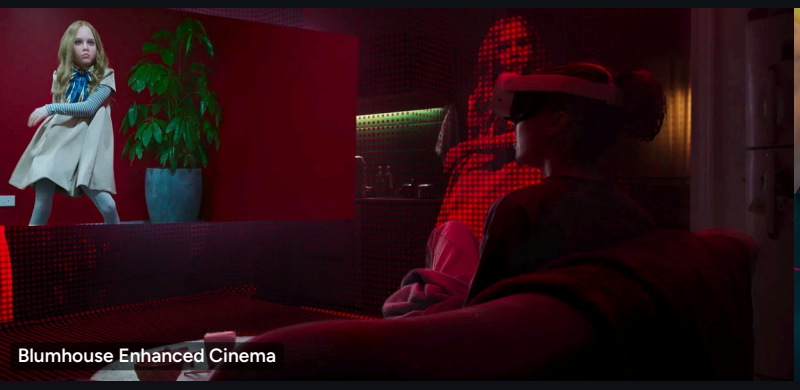
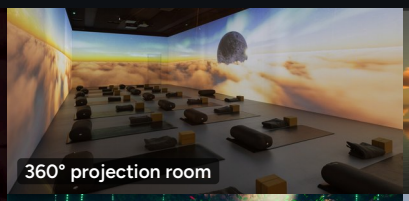
Charles Grassi

Graphics Programmer & Technical Artist

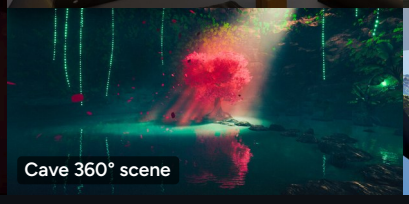
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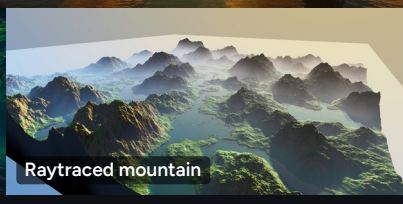
I've been drawing with mathematics for years, and since 2020 it has been my job.


Blumhouse Enhanced Cinema


360° projection room


360° projection room


Cave 360° scene


Raytraced mountain

EXPERIENCE

HYPERDRYFT Co-founder & Graphics Lead — 2026
A live-visuals engine on WebGPU: node graphs compile to multipass WGSL, driven by live audio.

Events Arena Chief Technology Officer — 2025–now
Built the company's booking platform and its AI agent.

Nexus Studios Technical Artist & Graphics Programmer — 2025
Worked on Blumhouse Enhanced Cinema for Meta Quest, nominated for a 2026 Primetime Emmy. Real-time global illumination for VR.

SABDA Chief Technology Officer — 2023–2024
Led the team behind a 360° projection installation, with real-time 3D across several servers.

Little Red Dog Games Lead Graphics Programmer — 2023
Led a small team on compute-shader and Burst VFX in Unity.

22cans Senior Graphics Programmer — 2023
A material shader library, plus grass and volumetric lighting for mobile and desktop.

Scapin' / The Sandbox Graphics Programmer — 2021–2022
Custom mobile lighting, stylized parallax clouds and screen-space god rays.

Ubisoft Technical Artist — 2020–2021
Shaders and VFX on a small title.

ARTICLES

Neural Shaders, Explained With Clouds — A small network trained against a path tracer.

GPU Cache Hierarchy — Why one texture fetch costs more than another.

WHAT I USE

Rendering
Real-time GI, PBR, volumetrics, compute, SDF, VFX, AR/VR

Languages and engines
C++, C#, Python, Rust, Unreal (Niagara), Unity (URP, HDRP)

AI and backend
Claude API, tool-use agents, MCP, FastAPI, PostgreSQL

APIs and shading
WebGPU/WGSL, DX11/12, Vulkan, Metal, OpenGL, HLSL, GLSL

Profiling
RenderDoc, NSight, PIX, Frame Debugger

Spoken
French (native), English (fluent)